Arij Asad

Doctoral Researcher in Applied Mathematics and Fluid Dynamics

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Research Profile

Doctoral researcher in applied mathematics and fluid dynamics at the University of Bristol, studying buoyancy-driven mixing and plume-generated stratification in confined environments, with applications to the storage of nuclear waste. Combines planar laser-induced fluorescence (PLIF) experiments with similarity analysis, mathematical modelling and finite-element simulations of confined and gravity-driven flows.

Research interests: buoyant plumes; mixing and stratification; experimental fluid dynamics; confined and gravity-driven flows; computational fluid dynamics; finite-element methods.

Education

Doctor of Philosophy (PhD); research in applied mathematics and fluid dynamics

School of Electrical, Electronic and Mechanical Engineering, University of Bristol | Nov 2022-present

- Thesis: *Mixing behaviour of plumes*.
- Supervisors: Dr Andrew Lawrie and Dr Stuart Thomson.
- Expected completion: October 2026.

BA (Cantab), Mathematics (Mathematical Tripos)

Murray Edwards College, University of Cambridge | Oct 2019-Jun 2022

- MA (Cantab), conferred Apr 2026.

Research Experience

Doctoral Researcher

University of Bristol | Nov 2022-present

- Investigate mixing and stratification generated by laminar, transitional and turbulent buoyant plumes in confined rectangular and circular geometries, with and without laminar crossflow, motivated by hydrogen ventilation in geological disposal facilities.
- Designed and conducted PLIF experiments to obtain time-resolved concentration and density fields for single-source, twin-source and ventilated configurations.
- Developed scaling laws and similarity solutions for stratified-layer growth and profile evolution, including the effects of plume regime, crossflow and enclosure geometry.
- Built finite-element simulations in FEniCS for incompressible flow and advection-diffusion and ran computational studies on Bristol's Slurm high-performance computing systems.

MSc Research Project Support

University of Bristol | Jan 2026-Sep 2026

Provided day-to-day research guidance to a student on *Stratification Development in a Confined Region: An Experimental Comparison of Density-Layer Development Generated by Single- and Double-Nozzle Buoyant Flows*.

Cambridge Mathematics Open Intern

Faculty of Mathematics, University of Cambridge | Jul-Aug 2022

Completed an eight-week research project on a Navier-slip ellipsoid in shear flow, supervised by Dr Catherine Kamal; presented the work at the Cambridge Mathematics Placements Presentation Day.

Manuscripts and Research Software

Manuscripts in preparation

- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *On the mixing behaviour of plumes in a confined environment*.
- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *Stratification development in a laminar crossflow*.
- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *Stratification development in a confined circular domain with and without laminar crossflow*.

Research software

- **PLIF acquisition and control software:** C client-server tools for synchronising laser pulsing and DALSA camera image capture across a host computer and BeagleBone Black.
- **MATLAB PLIF analysis workflow:** fluorescence calibration and image processing, density-field reconstruction and stratification analysis.

Talks and Presentations

Osborne Reynolds Day
On the mixing behaviour of plumes in a confined environment (poster)
Jun 2026

British Applied Mathematics Colloquium (BAMC), Norwich
On the mixing behaviour of plumes in a confined environment (poster)
Mar 2026

Nuclear Waste Services Research Support Office Annual Conference, Manchester
How does hydrogen build up in a closed vault environment? (poster)
Jan 2026

2nd European Fluid Dynamics Conference, Dublin
Mixing behaviour of plumes
Aug 2025

NWS RSO Annual Conference, Bristol
Hydrogen ventilation in a GDF (poster)
Jan 2025

NWS RSO Annual Conference, Sheffield
Mixing behaviour of laminar plumes with and without a ventilating crossflow
Jan 2024

NWS RSO Advanced Manufacturing Webinar (online)
Ventilation of hydrogen in UILW vaults (recording)
Oct 2023

NWS RSO Annual Conference, Leeds
Hydrogen ventilation in a geological disposal facility
Jan 2023

Cambridge Mathematics Placements Presentation Day
The motion of anisotropic carbon nanoparticles in shear flow: a hydrodynamic perspective
Aug 2022

Murray Edwards College Maths Circle, Cambridge
Sphere packing (talk for the Murray Edwards undergraduate cohort)
2019

Technical Skills

Experimental methods: PLIF acquisition and calibration; MATLAB image processing; extraction and analysis of time-resolved concentration and density fields.

Programming and computing: MATLAB, Python, C, Git/GitHub, Linux/Bash and LaTeX; Slurm HPC on Bristol's BlueCrystal and BluePebble systems.

Numerical methods: Scaling and similarity analysis; FEniCS finite-element simulation of incompressible flow and advection-diffusion.

Languages: Urdu (reading and conversational proficiency).

Teaching and Supervision

Undergraduate Mathematics Supervisor

University of Cambridge colleges | Oct 2024-present

- Delivered Part II Fluid Dynamics II supervisions to 29 students across ten colleges in 2024/25 and 2025/26.
- Mark written work and lead example-sheet and revision supervisions.
- Provide postgraduate references and academic progression support when requested.

Mathematics Tutor and Mock Interviewer

Dipont International | Oct 2025-present

- Provide admissions-test preparation, personal statement guidance and mock interviews for Oxford and Cambridge applicants.
- Delivered 33 hours of intensive small-group mathematics teaching and mock-interview preparation on a two-week Wuxi residential programme in Jul 2026, with structured feedback.

Graduate Teacher Level 1 and Demonstrator

University of Bristol | Sep 2023-Jan 2026

- Prepared and delivered weekly tutorials in Probability and Statistics and Introduction to Pure Mathematics as Graduate Teacher Level 1.
- Demonstrated Engineering Mathematics, Modelling in Applied Mathematics, Applied Linear Algebra and Continuum Mechanics.

Private Mathematics Tutor

Arij Asad Maths; primarily online, with occasional in-person teaching | Oct 2019-present

- Provide direct individual tuition in A-level Mathematics, Further Mathematics, Physics and Computer Science, together with undergraduate mathematics topics.
- Prepare candidates for STEP and TMUA (and MAT, 2020-2025) through diagnostic work, timed papers and Cambridge-style mock interviews.

Admissions Mathematics Tutor and Mock Interviewer

Blue Education; independent contractor | Nov 2019-present

Conduct Oxbridge Mathematics mock interviews and provide written feedback.

Academic Leadership

She Talks Science STEP Summer School

Murray Edwards College, University of Cambridge | Director, 2025; Deputy Director, 2023-2024; Tutor, 2019-2022

- As Director, led a four-day residential programme for 50 students; co-led the selection of 50 participants from 184 applicants; coordinated five tutors, three supervisors and three enrichment speakers.
- Led the academic programme, timetable, staff briefing and safeguarding; taught inequalities classes and Cambridge-style supervisions.
- As Deputy Director, contributed to teaching materials, classes, supervisions and residential activities.

Admissions, Examining and Olympiads

STEP Supervising Assessor

OCR (Oxford Cambridge and RSA) | Jun-Jul 2026

- Supervised marking of STEP Mathematics 2 Mechanics questions 9 and 10, providing real-time guidance to markers and checking scripts for consistent application of the mark scheme.
- Sampled marked scripts, documented quality checks and reviewed scripts requiring manual allocation before distribution to marking teams.

STEP Marker

OCR (Oxford Cambridge and RSA) | 2024-present

Marked STEP Mathematics 3 in 2024-2026 and STEP Mathematics 2 in 2025-2026, including Mechanics in 2026; applied mark schemes consistently and provided feedback on their implementation.

Mathematics Undergraduate Admissions Interviewer

Homerton College, University of Cambridge | Dec 2024

Interviewed and assessed Mathematics applicants during a four-day online admissions cycle as part of a two-person panel.

Assessment Associate (question vetting)

Pearson VUE | Oct-Nov 2024

Reviewed confidential mathematics assessment material for technical accuracy and clarity.

Undergraduate Mathematics Marker

University of Bristol | 2024-2025

Marked Applied Analysis B, Probability and Statistics, Applied Linear Algebra coursework, Engineering Mathematics and Introduction to Pure Mathematics.

UK Mathematical Olympiad for Girls Vetter and Marker

United Kingdom Mathematics Trust | 2019

Reviewed draft problems for mathematical correctness, scope, difficulty and clarity, and marked Olympiad scripts as part of the national marking team.

Outreach and Enrichment

Widening Participation Tutor

University of Bristol | Jul 2023-Oct 2026

Delivered university access sessions and research talks for school pupils aged 11-18 through Insight into Bristol, Next Step Bristol, Virtual Summer School and Sutton Trust Summer School programmes.

Admissions Talks and Webinars

2020-2026

- COMPOS interview-preparation webinar (through Blue Education), Aug 2026.
- Dipont International Oxbridge admissions talk, Shanghai, Jul 2026.
- Blue Education Oxbridge Interview Seminar, Oct 2025.
- Mathematics Open Day, University of Cambridge, 2022.
- Sutton Trust Experience Cambridge Day, 2020-2022.

Summer School Teaching

University of Cambridge | 2019-2022

Taught on the Sutton Trust and Intro to Maths at Uni summer schools.